in the flow. There is also a transitional stage between laminar and turbulent flows, in which the dye stream will wander about and show intermittent bursts of mixing, followed by a more laminar behavior.

The Reynolds number (*Re*), provides a useful way of characterizing the flow. It is defined as:

$$Re = \frac{vd}{\nu} \qquad (1)$$

where (ν) is the kinematic viscosity of the water (Figure 7.2), v is the mean flow velocity and d is the diameter of the pipe.

The Reynolds number is a dimensionless parameter that is the ratio of the inertial (destabilizing) force to the viscosity (stabilizing) force. As *Re* increases, the inertial force becomes relatively larger, and the flow destabilizes and becomes fully turbulent.

The Reynolds experiment determines the critical Reynolds number for pipe flow at which laminar flow (*Re<2000*) becomes transitional (*2000<Re<4000*) and the transitional flow becomes turbulent (*Re>4000*). The advantage of using a critical Reynolds number, instead of critical velocity, is that the results of the experiments are applicable to all Newtonian fluid flows in pipes with a circular cross-section.

Temperature (degree C)	Kinematic viscosity v (m^2/s)	Temperature (degree C)	Kinematic viscosity v (m^2/s)
0	1.793E-06	25	8.930E-07
1	1.732E-06	26	8.760E-07
2	1.674E-06	27	8.540E-07
3	1.619E-06	28	8.360E-07
4	1.522E-06	29	8.180E-07
5	1.520E-06	30	8.020E-07
6	1.474E-06	31	7.850E-07
7	1.429E-06	32	7.690E-07
8	1.386E-06	33	7.530E-07
9	1.346E-06	34	7.380E-07
10	1.307E-06	35	7.240E-07
11	1.270E-06	36	7.110E-07
12	1.235E-06	37	6.970E-07
13	1.201E-06	38	6.840E-07
14	1.169E-06	39	6.710E-07
15	1.138E-06	40	6.580E-07
16	1.108E-06	45	6.020E-07
17	1.080E-06	50	5.540E-07
18	1.053E-06	55	5.110E-07
19	1.027E-06	60	4.760E-07
20	1.002E-06	65	4.430E-07
21	9.780E-07	70	4.130E-07
22	9.550E-07	75	3.860E-07
23	9.330E-07	80	3.630E-07
24	9.110E-07	85	3.420E-07

Figure 7.2: Kinematic Viscosity of Water at Atmospheric Pressure.